Introduction to Machine Learning

Sample knowledge-base document for RAG ingestion and retrieval testing.

1. What Is Machine Learning?

Machine learning is a branch of artificial intelligence in which algorithms learn useful patterns from data and use those patterns to make predictions or decisions. Instead of writing every rule manually, developers provide examples and an algorithm learns from them.

Key idea: What Is Machine Learning? is an important foundational topic when learning and applying introduction to machine learning. Understanding the concept, its purpose, and common examples helps when building practical software systems.

2. Data and Features

A dataset contains examples used for learning. Features are input variables that describe each example. A target or label is the value the model is expected to predict in supervised learning.

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3. Supervised Learning

Supervised learning uses labeled examples. Regression predicts continuous numerical values such as house prices. Classification predicts categories such as spam versus not spam or fraudulent versus legitimate.

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4. Unsupervised Learning

Unsupervised learning works with data that does not have predefined target labels. Clustering can group similar customers or documents. Dimensionality reduction can represent complex datasets with fewer dimensions.

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5. Training and Testing

A model is trained using training data and evaluated using data it has not seen during training. A validation set can be used during development for model selection and tuning. The goal is good performance on unseen data.

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6. Overfitting and Underfitting

Overfitting happens when a model learns training data too closely, including noise, and then performs poorly on new data. Underfitting occurs when a model is too simple to capture important patterns.

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7. Common Algorithms

Common algorithms include linear regression, logistic regression, decision trees, random forests, support vector machines, k-nearest neighbors, naive Bayes, gradient boosting, k-means, and neural networks.

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8. Evaluation Metrics

Regression can use MAE, MSE, RMSE, and R-squared. Classification can use accuracy, precision, recall, F1 score, ROC-AUC, and confusion matrices. The correct metric depends on the business problem.

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9. Feature Engineering

Feature engineering transforms raw data into useful inputs for a model. Examples include extracting the day of week from a timestamp, encoding categories, scaling numerical values, and creating meaningful ratios.

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10. Machine Learning Workflow

A typical workflow includes collecting data, cleaning it, exploring it, preparing features, splitting datasets, selecting a model, training, evaluating, tuning, deploying, and monitoring. Data quality often matters as much as algorithm choice.

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